Journal of the Franklin Institute 363 (2026) 108380

Contents lists available at ScienceDirect

Journal of the Franklin Institute

journal homepage: www.elsevier.com/locate/fi

Sliding mode fault-tolerant control of quadrotor UAV based on adaptive disturbance estimator under delta operator framework

Shuo Gao [a], Bo-Chao Zheng [b,*], Hanqing Qu [b]

[a] School of Automation, Nanjing University of Information Science and Technology, 210044, Nanjing, Jiangsu, China
[b] CICAEET, School of Automation, Nanjing University of Information Science and Technology, 210044, Nanjing, Jiangsu, China

ARTICLE INFO

Keywords:
Quadrotor UAV
Delta operator
Adaptive disturbance estimator
Sliding mode control

ABSTRACT

To address the challenge of quadrotor UAVs failing to complete their tasks due to abrupt external disturbances and actuator failures, this paper proposes a nonlinear recursive sliding mode control (NRSMC) strategy integrated with an adaptive disturbance estimator (ADE) within the delta operator framework. First, an adaptive disturbance estimator is designed for the quadrotor UAV system in the delta operator framework. This estimator is capable of estimating both the total disturbance and its derivatives, which enhances the accuracy and adaptability of disturbance estimation, particularly in the presence of abrupt disturbances. Second, a nonlinear recursive sliding mode control strategy is designed. By combining the disturbance estimation results, the control law ensures the system converges quickly and smoothly. Finally, simulation experiments are conducted to validate the proposed method. The results show that the approach can effectively compensate for abrupt disturbances and actuator failures. This allows the quadrotor UAV to maintain stable trajectory tracking and successfully complete its tasks when facing abrupt disturbances and actuator failures.

1. Introduction

The quadrotor UAV(Unmanned Aerial Vehicle) has a simple structure, consisting of four symmetrically distributed motors and rotors. It can achieve precise three-dimensional motion by adjusting the rotor speeds. Due to its excellent flexibility and stability, quadcopter drones have gained an important status and are widely used in fields such as the military, agriculture, and industry. Recent studies have affirmed its advantages and emphasized their continuously expanding applications. Arabi Aliabad et al. [1] described a study focused on identifying and monitoring gardens in urban areas of arid regions using aerial and satellite imagery. Mendu et al. [2] described a systematic review of the application of unmanned aerial vehicles (UAVs) in power line inspections, highlighting current innovations, trends, and future prospects. However, when performing complex tasks, unmanned aerial vehicles (UAVs) are often susceptible to abrupt external disturbances (e.g. caused by strong winds or turbulent airflow) as well as system faults resulting from prolonged operation, such as actuator performance degradation or complete failure. These factors can severely compromise flight stability, leading to loss of control or even catastrophic crashes. Although there have been many studies on fault-tolerant control of UAVs based on sliding mode control, there are still two problems that need to be solved.

- Existing control methods remain inadequate in addressing abrupt disturbances. For example, some robust control methods may generate large instantaneous control actions and overshoot when encountering abrupt disturbances, which is caused by the instantaneous changes of the disturbances.

* Corresponding author.
E-mail address: zhengbochao81@126.com (B.-C. Zheng).

https://doi.org/10.1016/j.jfranklin.2025.108380
Received 30 July 2025; Received in revised form 22 December 2025; Accepted 30 December 2025
Available online 2 January 2026
0016-0032/© 2026 The Franklin Institute. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

- Most of the models used in existing methods focus on UAV attitude angle tracking control. Although a few models can handle both position and attitude tracking, they lack generalized models that include multiple types of faults and accurately describe system dynamics under delta operator framework.

To address the problems above, and inspired by the mentioned studies, this paper makes the following contributions:

- A quadrotor UAV model within the delta operator framework is constructed, which more accurately captures the system's dynamic performance under discrete control and is particularly suitable for digital controller implementation. Based on this model, an adaptive disturbance estimator capable of addressing abrupt disturbances is proposed.
- A nonlinear recursive sliding mode fault-tolerant control strategy is designed based on the proposed adaptive disturbance estimator, enabling the quadrotor UAV to accurately and smoothly track the desired trajectory despite actuator faults and abrupt disturbances. Compared to existing methods that consider only single-type faults, this scheme offers broader applicability and engineering practicality.

So far, researchers have invested a significant amount of effort in fault-tolerant sliding mode control for quadrotor UAVs. Mai et al. [3] presented a UAV model including actuator faults,and presented a robust fault-tolerant approach. Zhou et al. [4] presents a fault-tolerant control scheme addressing the translational and rotational dynamics of a fully-actuated quadrotor UAV system model. Several sliding mode control strategies based on disturbance estimators have been proposed, which enable fault-tolerant control for quadrotor UAVs, improve control accuracy, and reduce convergence time, thereby providing an effective solution for sliding mode fault-tolerant control of UAVs [4–6][16–19].

In order to estimate disturbances more accurately, various adaptive estimators have been proposed. Adaptive estimation of uncertainty bounds is used for compensation and fault separation [7,8]. Three types of adaptive disturbance estimators have been presented, providing a reference value for adaptive estimation [9–11]. Ranjan et al. [12] proposed a fixed-time differentiator to estimate outputs' derivatives, largely compensating for initial condition problems and solving the delay problem. These adaptive methods can accurately separate faults or estimate the impact of disturbances and faults on the system.

Delta operators are commonly used in control to convert between continuous and discrete systems. In practical applications, the state update of the quadrotor UAV is usually implemented in a discrete system. Most studies still use idealized continuous systems, taking discrete data points from continuous signals, but this approach cannot fully capture the system's dynamic behavior. To achieve more precise state updates and control in digital controllers, an appropriate discretization method is needed to convert between continuous and discrete systems. Traditional discrete methods, such as the z-transform and q-operator transformation, can suffer from accuracy degradation or even system instability at high sampling frequencies. In contrast, the Delta operator has stronger robustness and offers significant advantages in the consistency and numerical stability of continuous-discrete systems at high sampling frequencies. Yang et al. [13] proposes a disturbance estimator based on the Delta operator to address disturbance estimation and compensation problems in uncertain systems, extending to Multi-Input Multi-Output (MIMO) nonlinear systems. Ginoya et al. [14] utilizes the Delta operator for systems with parameter uncertainties and external nonlinear disturbances, proposing a sliding mode estimator and controller that improves the system's disturbance rejection capability.

The remainder of this paper is organized as follows. Section 2 establishes a model with disturbances and actuator faults within the delta operator framework. Section 3 proposes an adaptive disturbance estimator. Sections 4 and 5 elaborate on the proposed nonlinear recursive sliding mode control strategy, including the attitude controller and the position controller. Sections 6 and 7 present simulation results and discussions. Finally, Section 8 concludes the paper and suggests directions for future research.

2. UAV model under the delta operator framework

Following the model in [3], considering actuator failures and abrupt external disturbances, the kinematic model of the UAV in a continuous-time framework can be given by

$$\begin{cases} \ddot{x} = \dfrac{(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi)(\rho_1 u_1 - f_1) - k_x\dot{x}}{m} + d_1, \\ \ddot{y} = \dfrac{(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi)(\rho_1 u_1 - f_1) - k_y\dot{y}}{m} + d_2, \\ \ddot{z} = \dfrac{(\cos\phi\cos\theta)(\rho_1 u_1 - f_1) - mg - k_z\dot{z}}{m} + d_3, \\ \ddot{\phi} = \dfrac{l}{I_{xx}}((\rho_2 u_2 - f_2) - k_\phi\dot{\phi}) + d_4, \\ \ddot{\theta} = \dfrac{l}{I_{yy}}((\rho_3 u_3 - f_3) - k_\theta\dot{\theta}) + d_5, \\ \ddot{\psi} = \dfrac{l}{I_{zz}}((\rho_4 u_4 - f_4) - k_\psi\dot{\psi}) + d_6. \end{cases} \tag{1}$$

where u_1, u_2, u_3, u_4 are the control inputs for altitude, roll, pitch, and yaw respectively, l is the distance from each rotor to the center of the UAV, $x, y, z, \phi, \theta, \psi$ are the position and attitude angles of the UAV, $d_i, i = 1, 2 \ldots, 6$ are unmodeled disturbances, including external disturbances and state coupling effects, $\rho_i, i = 1, 2, 3, 4$ represent the remaining effectiveness factors, which characterize the impact of multiplicative faults caused by partial loss of effectiveness in the actuators, $f_i, i = 1, 2, 3, 4$ represent additive faults induced by actuator bias, $k_i, i = 1, 2, \ldots, 6$ are the aerodynamic drag coefficients, and I_{xx}, I_{yy}, I_{zz} are the moments of inertia.

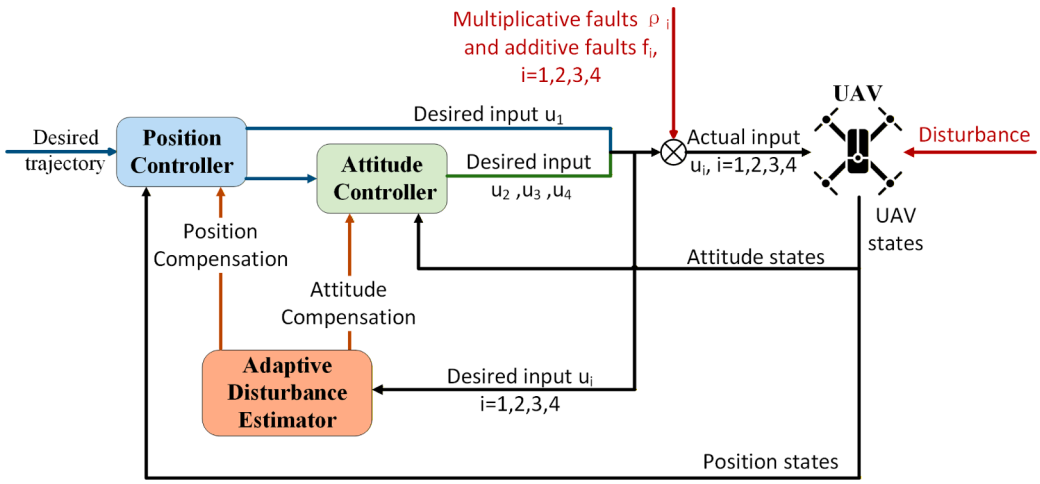

Fig. 1. Control architecture of the quadrotor UAV.

The physical meaning of the state vector and control inputs varies across different degrees of freedom. To present the system structure more clearly, the state equations of the continuous-time system are divided into two parts: the position dynamics and the attitude dynamics. The state-space model of the UAV in the continuous-time framework is given by

$$\begin{cases} \dot{x}_i = \bar{A}_i x_i + \bar{B}_i b_i u_1 - f_1 + \bar{F}_i d_i, & i = 1,2,3, \\ \dot{x}_i = \bar{A}_i x_i + \bar{B}_i b_i u_{i-2} - f_{i-2} + \bar{F}_i d_i, & i = 4,5,6. \end{cases} \tag{2}$$

where $x_1 = \begin{bmatrix} x \\ \dot{x} \end{bmatrix}$, $x_2 = \begin{bmatrix} y \\ \dot{y} \end{bmatrix}$, $x_3 = \begin{bmatrix} z \\ \dot{z} \end{bmatrix}$, $x_4 = \begin{bmatrix} \phi \\ \dot{\phi} \end{bmatrix}$, $x_5 = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}$, $x_6 = \begin{bmatrix} \psi \\ \dot{\psi} \end{bmatrix}$, $\bar{A}_1 = \begin{bmatrix} 0 & 1 \\ 0 & -\frac{k_x}{m} \end{bmatrix}$, $\bar{A}_2 = \begin{bmatrix} 0 & 1 \\ 0 & \frac{k_y}{m} \end{bmatrix}$, $\bar{A}_3 = \begin{bmatrix} 0 & 1 \\ 0 & \frac{k_z}{m} \end{bmatrix}$, $\bar{A}_4 = \begin{bmatrix} 0 & 1 \\ 0 & \frac{lk_\phi}{I_{xx}} \end{bmatrix}$, $\bar{A}_5 = \begin{bmatrix} 0 & 1 \\ 0 & \frac{lk_\theta}{I_{yy}} \end{bmatrix}$, $\bar{A}_6 = \begin{bmatrix} 0 & 1 \\ 0 & \frac{lk_\psi}{I_{zz}} \end{bmatrix}$, $\bar{B}_i = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $\bar{F}_i = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $b_1 = \frac{\rho_1(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi)}{m}$, $b_2 = \frac{\rho_1(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi)}{m}$, $b_3 = \frac{\rho_1\cos\phi\cos\theta}{m}$, $b_4 = \frac{\rho_2 l}{I_{xx}}$, $b_5 = \frac{\rho_3 l}{I_{yy}}$, $b_6 = \frac{\rho_4 l}{I_{xx}}$.

In practical applications, the state update of the UAV is implemented in a discrete-time system. In fault-tolerant control of the UAV, the system is required to exhibit strong robustness and real-time responsiveness to handle circumstances such as actuator failures and external disturbances. However, continuous-time modeling may fail to fully capture the system's dynamic characteristics, potentially affecting both robustness and response speed. Therefore, it is necessary to discretize (2) using an appropriate approach. Traditional discretization approaches, such as the z-transform and q-operator transform, can achieve discretization but often lead to numerical errors and even system instability under high sampling frequencies. In contrast, the delta operator exhibits better consistency between continuous-time and discrete-time systems, as well as improved numerical stability at high sampling rates. Thus, the UAV system in this paper is modeled within the delta operator framework.

The definition of the delta operator is given by

$$\delta z_k = \begin{cases} \frac{d}{dt} z(t), & \text{if } T = 0, \\ \frac{z_{k+1} - z_k}{T}, & \text{if } T \neq 0. \end{cases} \tag{3}$$

where $z_k = z(kT)$ is the sampled value of the continuous-time function $z(t)$ at the kth sampling instant, T is a small sampling period used to ensure accurate discrete-time modeling. According to the definition, when $T = 0$, the system behaves as a continuous-time system; whereas when $T \neq 0$, the system behaves as a discrere time system.

According to Reference [15], the continuous-time state equation of the UAV can be represented in the delta operator framework as

$$\begin{cases} \delta x_i = A x_i + B_i b_i u_i + \Theta_i, & i = 1,2,3, \\ \delta x_i = A x_i + B_i b_i u_{i-2} + \Theta_i, & i = 4,5,6. \end{cases} \tag{4}$$

where $A_i = \frac{e^{\bar{A}_i T} - I}{T}$, $B_i = \frac{1}{T}\int_0^T e^{\bar{A}_i T}\bar{B}_i\, dt$, $\Theta_i = B_i f_i + F_i d_i$, and $F_i = \frac{1}{T}\int_0^T e^{\bar{A}_i T}\bar{F}_i\, dt$.

The block diagram of the overall control architecture is shown in Fig. 1.

3. Adaptive disturbance estimator

In UAV systems, modeling errors, external disturbances, and actuator failures often reduce control accuracy and system robustness. To address these challenges, disturbance estimators are used to compensate for uncertainties, improving both trajectory tracking and disturbance rejection. Traditional estimation methods, such as conventional disturbance observers, can handle slow-varying disturbances like light wind. However, their accuracy tends to decrease and may even lead to instability when faced with abrupt disturbances like strong gusts. To improve estimation accuracy and enhance adaptability to abrupt disturbances, we design a disturbance estimator that estimates the disturbance itself and its derivatives up to the rth order. The estimator uses a recursive structure, where higher-order estimates progressively refine the updates of lower-order terms. Additionally, we introduce an adaptive estimator gain that dynamically adjusts the estimator gains. This helps to speed up convergence and improve disturbance rejection performance.

Lemma 1 ([15]). *Under the delta operator framework, we sample continuous-time functions $x(t)$, $y(t)$ with a fast sampling period T. At the kth time step, we use $x_k = x(kT)$, $y_k = y(kT)$ to denote the sampled values of $x(t)$, $y(t)$. The delta of their product is:*

$$\delta(x_k y_k) = (\delta x_k) y_k + x_k(\delta y_k) + T(\delta x_k)(\delta y_k). \tag{5}$$

Based on the delta operator state (3), we construct a disturbance estimator for the UAV system.

As the first step, we estimate the disturbance. The estimated value of the disturbance is:

$$\widehat{\Theta}_i = L_{i1} x_i + \sigma_i^1. \tag{6}$$

where $L_{i1} \in \mathbb{R}^{2\times 2}$ are estimator gains, and all its elements are positive, σ_i^1 is an auxiliary variable. The update rule of σ_i^1 is given by

$$\delta\sigma_i^1 = -L_{i1}(Ax_i + BU + \widehat{\Theta}_i) \tag{7}$$

where $U = \begin{cases} u_i, & i = 1,2,3 \\ u_{i-2}, & i = 4,5,6 \end{cases}$,

However, fixed-gain disturbance estimators often cannot effectively handle abrupt disturbances, which can result in increased estimation errors and may even compromise system stability. To enhance the system's adaptability, this paper designs an adaptive gain that can adjust itself in response to changes in disturbance characteristics.

We design an adaptive estimator gain L_i^j, where $i = 1,2,\ldots,6$ represents the six degrees of freedom of the UAV; $j = 1,2,\ldots,r$ represents the estimation of the disturbance and its derivatives up to the $r-1$ order.

The update rule for the adaptive gain is:

$$\delta L_i^j = R_i^j \operatorname{diag}\left\{\gamma_i^j, \gamma_i^j\right\} - (L_i^{j0} - L_i^j)\operatorname{diag}\left\{\lambda_i^j, \lambda_i^j\right\} \tag{8}$$

where $L_i^j \in \mathbb{R}^{2\times 2}$ are adaptive estimator gains, $L_i^{j0} \in \mathbb{R}^{2\times 2}$ are the initial value of L_i^j, and all its elements are positive, $\gamma_i^j > 0$, $\lambda_i^j > 0$ are parameters, $R_{ij} = \begin{bmatrix} \delta\widehat{\delta^j\Theta}_{i1}\,\|\mathrm{sgn}(\widehat{\delta^j\Theta}_{i1})\| & 0 \\ 0 & \delta\widehat{\delta^j\Theta}_{i2}\,\|\mathrm{sgn}(\widehat{\delta^j\Theta}_{i2}\| \end{bmatrix}$, $\widehat{\delta^j\Theta}_i = \begin{bmatrix} \widehat{\delta^j\Theta}_{i1} \\ \widehat{\delta^j\Theta}_{i2} \end{bmatrix}$ is the estimated value of $\delta^j\Theta_i$, and $\delta^j\Theta_i$ is the jth derivative of the disturbance.

Then, the (6) can be rewritten as:

$$\widehat{\Theta}_i = L_i^1 x_i + \sigma_i^1. \tag{9}$$

To capture the variation trend of the disturbance and improve estimation accuracy, we introduce $\widehat{\delta\Theta}_i$ into σ_i^1 to adjust $\widehat{\Theta}_i$. Thus, the (7) can be rewritten as:

$$\delta\sigma_i^1 = -(L_i^1 + T\delta L_i^1)(A_i x_i + B_i U + \widehat{\Theta}_i) + \widehat{\delta\Theta}_i - \delta L_i^1 x_i. \tag{10}$$

According to the (9),using the delta operation on $\widehat{\Theta}_i$, we can obtain

$$\delta\widehat{\Theta}_i = \delta(L_i^1 x_i) + \delta\sigma_i^1. \tag{11}$$

By Lemma 1, substituting (4) and (10) into (11), we can obtain:

$$\delta\widehat{\Theta}_i = (L_i^1 + T\delta L_i^1)\widetilde{\Theta}_i + \widehat{\delta\Theta}_i. \tag{12}$$

where $\widetilde{\Theta}_i = \Theta_i - \widehat{\Theta}_i$ is the estimation error of the disturbance. Using the delta operation on $\widetilde{\Theta}_i$, we can obtain

$$\begin{aligned} \delta\widetilde{\Theta}_i &= \delta\Theta_i - (L_i^1 + T\delta L_i^1)\widetilde{\Theta}_i - \widehat{\delta\Theta}_i \\ &= -(L_i^1 + T\delta L_i^1)\widetilde{\Theta}_i + \widetilde{\delta\Theta}_i. \end{aligned} \tag{13}$$

where $\widetilde{\delta\Theta}_i = \delta\Theta_i - \widehat{\delta\Theta}_i$ is the estimation error of the disturbance derivative.

As the second step, we estimate the disturbance derivative. The estimated value of $\delta\Theta_i$ is:

$$\widehat{\delta\Theta}_i = L_i^2 x_i + \sigma_i^2. \tag{14}$$

where σ_i^2 is an auxiliary variable. To capture the variation trend of the disturbance and improve estimation accuracy, we introduce $\widehat{\delta^2\Theta}_i$ into σ_i^2 to adjust $\widehat{\delta\Theta}_i$. The update rule of σ_i^2 is given by

$$\delta\sigma_i^2 = -(L_i^2 + T\delta L_i^2)(A_i x_i + B_i U + \widehat{\Theta}_i) + \widehat{\delta^2\Theta}_i - \delta L_i^2 x_i. \tag{15}$$

According to the (14), we can obtain

$$\delta\widehat{\delta\Theta}_i = \delta L_i^2 \delta x_i + \delta\sigma_i^2. \tag{16}$$

By Lemma 1, substituting (4) and (15) into (16), we can obtain:

$$\delta\widehat{\delta\Theta}_i = (L_i^2 + T\delta L_i^2)\widetilde{\Theta}_i + \widehat{\delta^2\Theta}_i. \tag{17}$$

Therefore, using the delta operation on $\widetilde{\delta\Theta}_i$,we can obtain

$$\delta\widetilde{\delta\Theta}_i = -(L_i^2 + T\delta L_i^2)\widetilde{\Theta}_i + \widetilde{\delta^2\Theta}_i. \tag{18}$$

where $\widetilde{\delta^2\Theta}_i = \delta^2\Theta_i - \widehat{\delta^2\Theta}_i$ is the estimation error of the 2-rd disturbance derivative. This process continues iteratively up to step $r-1$.

When $j = 3, \dots, r-1$,the estimated value of $\delta^{(j-1)}\Theta_i$ is:

$$\widehat{\delta^{(j-1)}\Theta}_i = L_i^j x_i + \sigma_i^j. \tag{19}$$

where σ_i^j is an auxiliary variable. We introduce $\widehat{\delta^j\Theta}_i$ into σ_i^j to adjust $\widehat{\delta^{(j-1)}\Theta}_i$. The update rule of σ_i^j is given by

$$\delta\sigma_i^j = -(L_i^j + T\delta L_i^j)(A_i x_i + B_i U + \widehat{\Theta}_i) + \widehat{\delta^j\Theta}_i - \delta L_i^j x_i. \tag{20}$$

Therefore, using the delta operation on $\widetilde{\delta^{(j-1)}\Theta}_i$, we can obtain

$$\begin{aligned}\delta\widetilde{\delta^{(j-1)}\Theta}_i &= \delta^j\Theta_i - \delta\widehat{\delta^{(j-1)}\Theta}_i \\ &= -(L_i^j + T\delta L_i^j)\widetilde{\Theta}_i + \widetilde{\delta^j\Theta}_i.\end{aligned} \tag{21}$$

When $j = r$, the estimated value of $\delta^{(r-1)}\Theta_i$ is:

$$\widehat{\delta^{(r-1)}\Theta}_i = L_i^r x_i + \sigma_i^r. \tag{22}$$

where σ_i^r is an auxiliary variable. The delta of σ_i^r is:

$$\delta\sigma_i^r = -(L_i^r + T\delta L_i^r)(A_i x_i + B_i U + \widehat{\Theta}_i) - \delta L_i^r x_i. \tag{23}$$

Therefore, using the delta operation on $\widetilde{\delta^{(r-1)}\Theta}_i$,we can obtain

$$\delta\widetilde{\delta^{(r-1)}\Theta}_i = -(L_i^r + T\delta L_i^r)\widetilde{\Theta}_i + \delta^r\Theta_i. \tag{24}$$

Up to this point, the adaptive estimator in this paper, based on (9), (14), (19), and (22), estimates the disturbance and its higher-order derivatives. At each level, the estimated value is used to correct the previous-order derivative. This can improve the system's adaptability to abrupt disturbances and increases estimation accuracy. Finally, the corrected estimated value of the disturbance is provided to the controller.

We define the error vector as:

$$\eta_i = \begin{bmatrix} \widetilde{\Theta}_i^\top & \widetilde{\delta\Theta}_i^\top & \cdots & \widetilde{\delta^{(r-2)}\Theta}_i^\top & \widetilde{\delta^{(r-1)}\Theta}_i^\top \end{bmatrix}$$

According to (13), (18), (21), and (24), the delta of η_i is:

$$\delta\eta_i = A_{\eta i}\eta_i + B_{\eta i}\delta^r\Theta_i. \tag{25}$$

where

$$A_{\eta_i} = \begin{bmatrix} -(L_i^1 + T\delta L_i^1) & I & 0 & \cdots & 0 \\ -(L_i^2 + T\delta L_i^2) & 0 & I & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -(L_i^{r-1} + T\delta L_i^{r-1}) & 0 & 0 & \cdots & I \\ -(L_i^r + T\delta L_i^r) & 0 & 0 & \cdots & 0 \end{bmatrix}, B_{\eta_i} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ I \end{bmatrix}$$

.

Theorem 1. *When $|\lambda_m| < \|P_{\eta i}B_{\eta i}\|$, the estimation error of the adaptive disturbance estimator designed in this paper is convergent, and its maximum value is $\dfrac{2\|P_{\eta i}B_{\eta i}\|\mu}{|\lambda_m|}$, where λ_m is the largest eigenvalue of $P_{\eta i}A_{\eta i} + A_{\eta i}^\top P_{\eta i}$, $|\lambda_m|$ is the absolute value of λ_m, μ is the maximum value of $\delta^r\Theta_i$, and $P_{\eta i}$ a positive definite matrix.*

Proof. From the definition of the delta operator, we can obtain: $L_i^j((k+1)T)$

$$\begin{aligned}
&= L_i^j(kT) + TR_i^j(kT)\,\mathrm{diag}\{\gamma_i^j, \gamma_i^j\} \\
&\quad - T(L_i^j(kT) - L_i^{j0})\,\mathrm{diag}\{\lambda_i^j, \lambda_i^j\} \\
&= (I - T\,\mathrm{diag}\{\lambda_i^j, \lambda_i^j\})L_i^j(k) + L_i^{j0}\,\mathrm{diag}\{\lambda_i^j, \lambda_i^j\} \\
&\quad + TR_i^j(kT)\,\mathrm{diag}\{\gamma_i^j, \gamma_i^j\}.
\end{aligned} \tag{26}$$

where $L_i^j(kT)$ and $R_i^j(kT)$ represent the sampled values at the kth time step of L_i^j and R_i^j respectively. Since all the elements in L_i^{j0} are positive, when $T\lambda_i^j < 1$, the diagonal elements of L_i^j are always positive, and the diagonal elements of $-(L_i^j + T\delta L_i^j)$ are always negative. The first column of $A_{\eta i}$ contains all negative elements, and its characteristic polynomial is:

$$\lambda^{2r} + c_1\lambda^{2r-1} + c_2\lambda^{2r-2} + \cdots + c_{2r}. \tag{27}$$

where $c_i, i = 1, 2, \ldots, 2r$ is the parameter formed by the elements of L_i^j. Thus, by appropriately configuring L_i^{j0}, through methods like pole placement, the eigenvalues of $A_{\eta i}$ can be placed within a circle centered at $\left(-\frac{1}{T}, 0\right)$ with radius $\frac{1}{T}$. Properly configuring γ_i^j and λ_i^j ensures that the elements in L_i^j remain within a reasonable range near their initial values, which guarantees system stability. Thus, there exists a positive definite matrix $P_{\eta i}$ that ensures $P_{\eta i}A_{\eta i} + A_{\eta i}^{\top}P_{\eta i}$ is negative.

Let the Lyapunov function be $V_1 = \eta_i^T P_{\eta i}\eta_i$. By Lemma 1, using the delta operation on V_1, we can obtain

$$\begin{aligned}
\delta V_1 &= \eta_i^{\top}P_{\eta i}\delta\eta_i + T(\delta\eta_i)^{\top}P_{\eta i}\delta\eta_i + (\delta\eta_i)^{\top}P_{\eta i}\eta_i \\
&= \eta_i^{\top}(P_{\eta i}A_{\eta i} + A_{\eta i}^{\top}P_{\eta i})\eta_i + 2\eta_i^{\top}M\delta^r\Theta_i \\
&\quad + (\delta^r\Theta_i)^{\top}M^{\top}\eta_i + (\delta^r\Theta_i)^{\top}N\delta^r\Theta_i \\
&\le \|\eta_i\|^2\lambda_m + 2\|\eta_i\|\mu\|M\| + \|N\|\mu^2 \\
&= \mu^2\left(\lambda_m\chi^2 + 2\|M\|\chi\right) + \|N\|.
\end{aligned} \tag{28}$$

where $M = P_{\eta i}B_{\eta i} + TA_{\eta i}^{\top}P_{\eta i}B_{\eta i}$, $N = TB_{\eta i}^{\top}P_{\eta i}B_{\eta i}$, $\chi = \dfrac{\|\eta_i\|}{\mu}$.

Since $\mu^2 > 0$, $\lambda_m < 0$, when T is sufficiently small, according to the properties of the quadratic equation, as long as $|\lambda_m| < 2\|P_{\eta i}B_{\eta i}\|$, it satisfies the condition in the parentheses; the quadratic equation will always be less than zero. In this case, $\delta V_1 < 0$. The error is limited to the maximum value $\|\eta_i\|_{max}$, and $\|\eta_i\|_{max}$ satisfies: $\lambda_m\|\eta_i\|_{max} + 2\|P_{\eta i}B_{\eta i}\| = 0$. That is:

$$\|\eta_i\|_{max} = \frac{2\|P_{\eta i}B_{\eta i}\|\mu}{|\lambda_m|}. \tag{29}$$

Thus, the proof of Theorem 1 is completed. □

Remark 1. Increasing the parameters in L_i^{j0} raises the initial gain, improving sensitivity but risking overshoot and high-frequency noise. Increasing γ_i^j accelerates gain adaptation at the cost of potential oscillations or instability. Increasing λ_i^j improves robustness and prevents gain drift, but excessive values reduce adaptive flexibility. Therefore, L_i^{j0} and γ_i^j must balance estimation quality and speed, while λ_i^j should be set according to robustness requirements.

4. Attitude controller

To improve the system's robustness and convergence speed, we design a two-level recursive sliding surface structure. By recursively constructing a second-level sliding surface within the first, the system achieves better dynamic performance and higher precision.

In the first-level sliding surface, we introduce a nonlinear function fal(·). This function accelerates convergence when the error is large and ensures smooth convergence when the error is small. It is defined as follows:

$$\mathrm{fal}(x, \alpha_i, \epsilon) = \begin{cases} \dfrac{x}{\epsilon^{1-\alpha_i}}, & |x| \le \epsilon \\ |x|^{\alpha_i}\mathrm{sgn}(x), & |x| > \epsilon \end{cases}. \tag{30}$$

where $0 < \alpha_i < 1$, for simplicity, we denote $\mathrm{fal}(x, \alpha_i, \epsilon)$ by $\mathrm{fal}(x)$.

We design the attitude controller based on the second row of (4), where $i = 4, 5, 6$. The two-level sliding surfaces can be given by:

$$s_{i1} = \delta e_i + C_{i1}e_i + C_{i2}\mathrm{fal}(e_i), \tag{31}$$

$$s_{i2} = s_{i1} + TC_{i3}\sum_{i=0}^{k-1}|s_{i1}|^{\beta_i}\,\mathrm{sgn}(s_{i1}). \tag{32}$$

where $C_{i1} > 0, C_{i2} > 0, C_{i3} > 0, \beta_i > 0$, $e_i = x_i - x_{id}$ represents the error between the actual attitude angle x_i and desired attitude angle x_{id}.

Remark 2. In the designed two-level sliding surfaces, the first-level sliding surface is designed based on the error e_i, enabling e_i to converge to zero, and uses the nonlinear function fal(·) to effectively smooth error dynamics. The second-level sliding surface is designed based on the first-level surface, ensuring rapid convergence of the first-level sliding surface. With the synchronous action of the two sliding surfaces, e_i can quickly converge to zero. Moreover, to further reduce chattering, the sgn function sgn(x) is replaced by $\tanh\left(\frac{x}{\epsilon}\right)$, where ϵ is a small value. For simplicity, $\tanh\left(\frac{x}{\epsilon}\right)$ is denoted by $\tanh(x)$.

Then, (30) can be rewritten as:

$$\mathrm{fal}(x,\alpha,\epsilon)=\begin{cases}\dfrac{x}{\epsilon^{1-\alpha}}, & |x|\le\epsilon\\ |x|^{\alpha}\tanh(x), & |x|>\epsilon\end{cases}. \tag{33}$$

(32) can be rewritten as:

$$s_{i2}=s_{i1}+TC_{i3}\sum_{i=0}^{k-1}|s_{i1}|^{\beta_i}\tanh(s_{i1}). \tag{34}$$

The control law is given by:

$$\begin{aligned}u_{i-2}=&\frac{1}{B_{i2}}(-C_{i1}\delta e_i-C_{i2}\tau_i-C_{i3}|s_{i1}|^{\beta_i}\tanh(s_{i1})\\&-\Gamma_{i1}s_{i2}-\Gamma_{i2}\tanh(s_{i2}))-\widehat{\Theta}_i).\end{aligned} \tag{35}$$

where $\tau_i=\begin{cases}\alpha_i|e_i|^{\alpha_i-1}\tanh(e_i)\delta e_i, & |e_i|\le\epsilon\\ \frac{C_{i2}}{\epsilon^{\alpha_i-1}}\delta e_i, & |e_i|>\epsilon\end{cases}$, $\Gamma_{i1}>0,\Gamma_{i2}>0$. Substituting (31) and (35) into equation (34), we have:

$$\delta s_{i2}=-\Gamma_{i1}s_{i2}-\Gamma_{i2}\tanh(s_{i2})+\widehat{\Theta}_i. \tag{36}$$

Theorem 2. *For the UAV system in (4), the proposed attitude controller given in (31)–(35) ensures that the tracking error e_i converges to zero.*

Proof. For the second-level sliding surface, let the Lyapunov function be $V_{s2}=s_{i2}^2$. From Lemma 1, we obtain:

$$\begin{aligned}\delta V_{s2}&=2s_{i2}\delta s_{i2}+T(\delta s_{i2})^2\\&=2s_{i2}\left(-\Gamma_{i1}s_{i2}-\Gamma_{i2}\tanh(s_{i2})+\widetilde{\Theta}_i\right)+T(\delta s_{i2})^2\\&=-2\Gamma_{i1}s_{i2}^2-2\Gamma_{i2}|s_{i2}|+2\widetilde{\Theta}_i s_{i2}+\frac{T}{2}(\delta s_{i2})^2\\&\le-2\Gamma_{i1}s_{i2}^2+2(\widetilde{\Theta}_i-\Gamma_{i2})|s_{i2}|+\frac{T}{2}(\delta s_{i2})^2.\end{aligned} \tag{37}$$

It can be seen that by appropriately selecting the values of Γ_{i1} and Γ_{i2}, we can ensure that $|\delta s_{i2}|$ has a small upper bound. As the sampling period T approaches zero, $T(\delta s_{i2})^2$ becomes a minimal value. In this case, as long as $\widetilde{\Theta}_i-\Gamma_{i2}<0$, the system will converge. Therefore, by appropriately selecting Γ_{i1}, Γ_{i2}, and T, we can make the second-level sliding surface $|s_{i2}|$ approximately converge to a constant value, which is caused by the accumulation on the first-level sliding surface.

For the first-level sliding surface, let the Lyapunov function be $V_{s1}=s_{i1}^2$. From (34), when s_{i2} converges to a constant value, we can obtain

$$\begin{aligned}\delta V_{s1}&=2s_{i1}\delta s_{i1}+T(\delta s_{i1})^2\\&=-2C_{i3}|s_{i1}|^{\beta_i+1}+TC_{i3}^2|s_{i1}|^{2\beta_i}\\&=-2|s_{i1}|^{\beta_i+1}C_{i3}\left(1-\frac{T}{2}C_{i3}|s_{i1}|^{\beta_i-1}\right).\end{aligned} \tag{38}$$

Since $|s_{i1}|^{\beta_i+1}>0$, when $0<\beta_i<1$, we can obtain that $|s_{i1}|^{1-\beta_i}>\dfrac{TC_{i3}}{2}$. Thus, when the sampling period T is sufficiently small, s_{i1} gradually converges to zero. In this case, $\delta e_i=-C_{i1}e_i-C_{i2}e_i|e_i|^{\alpha_i}\tanh\left(\frac{e_i}{\epsilon}\right)$

For the tracking error, let the Lyapunov function be $V_e=e_i^2$. Using the delta operation on V_e, we can obtain

$$\begin{aligned}\delta V_e&=2e_i\delta e_i+T(\delta e_i)^2\\&=e_i^{2\alpha_i}\left((-2+TC_{i1}^2)\mathrm{E}_i^2+(TC_{i1}C_{i2}-2C_{i2})\mathrm{E}_i+TC_{i2}^2\right).\end{aligned} \tag{39}$$

where $\mathrm{E}_i=|e_i|^{1-\alpha_i}$.The expression inside the parentheses can be regarded as a downward-opening linear quadratic equation in E_i, which has no solution within its domain.

Therefore, $\delta V_e<0$. In summary, the two-level recursive sliding surface designed in this paper is stable and ensures that the tracking error e_i converges to zero. Thus, the proof of Theorem 2 is complete. □

5. Position controller

For the first row of (4), the position controller is designed for the UAV. The position controller has $i = 1, 2, 3$, where the position error $\xi_i = x_i - x_{id}$ is defined. The variables represent the three position states of the UAV: x_{id} is the desired position of the UAV, and x_i is the position of the UAV, as per the attitude controller design. The design part is as follows, with two-level sliding surfaces:

$$\rho_{i1} = \delta\xi_i + C_{i1}\xi_i + C_{i2}\mathrm{fal}(\xi_i), \tag{40}$$

$$\rho_{i2} = \rho_{i1} + TC_{i3}\sum_{i=0}^{k-1}|\rho_i|^{\beta_i}\tanh(\xi_i), \tag{41}$$

where $C_{i1} > 0$, $C_{i2} > 0$, $C_{i3} > 0$, $\beta_i > 0$. The control law is given by:

$$\begin{aligned} U_i = &-C_{i1}\xi_i - C_{i2}\kappa_i - C_{i3}|\rho_i|^{\beta_i}\tanh(\xi_i) \\ &- \Gamma_{i1}\rho_2 - \Gamma_{i2}\tanh(\rho_2) - \widehat{\Theta}_i, \end{aligned} \tag{42}$$

where $\kappa_i = \begin{cases} \alpha_i|\xi_i|^{\alpha_i-1}\tanh(\xi_i)\delta\xi_i) & |\xi_i| \le \epsilon \\ \dfrac{C_{i2}}{\epsilon^{\alpha_i-1}}\delta\xi_i & |\xi_i| > \epsilon \end{cases}, \Gamma_{i1} > 0, \Gamma_{i2} > 0.$

Remark 3. Increasing the variables C_{i1}, C_{i2} and C_{i3} improves response speed, tracking accuracy, and nonlinear error handling respectively, yet excessive C_{i1} or C_{i2} may cause overshoot or oscillation. Adjusting α_i modifies sensitivity to large versus small errors. Increasing β_i enhances large-error sensitivity and accuracy but risks instability. Increasing Γ_{i1} and Γ_{i2} boosts response speed and disturbance rejection, though overly large values may induce oscillation or aggressive control.

Theorem 3. *Let $\psi_d = 0$, the position controller provides feedback to the attitude control system, with the expected angle and the input to the UAV system as follows:*

$$u_1 = m\sqrt{(U_1)^2 + (U_2)^2 + (U_3 + g)^2}, \tag{43}$$

$$\phi_d = \arcsin\left(\frac{-mU_2}{u_1}\right), \tag{44}$$

$$\theta_d = \arctan\left(\frac{U_1}{U_3 + g}\right). \tag{45}$$

Proof. From (1), it can be seen that the position control of the quadrotor UAV is carried out by computing the required accelerations through the total thrust u_1 and the attitude angles. These accelerations are then updated. By using the virtual control law obtained from (42), the desired accelerations for the three position components can be calculated as follows:

$$\begin{cases} U_1 = \dfrac{(\cos\phi\sin\theta\cos\psi + \sin\phi\sin\psi)u_1}{m} \\ U_2 = \dfrac{(\cos\phi\sin\theta\sin\psi - \sin\phi\cos\psi)u_1}{m} \\ U_3 = \dfrac{(\cos\phi\cos\theta)u_1}{m} - g \end{cases} \tag{46}$$

When $\psi_d = 0$, (46) can be written as:

$$\begin{cases} U_1 = \dfrac{\cos\phi\sin\theta u_1}{m} \\ U_2 = \dfrac{-\sin\phi u_1}{m} \\ U_3 + g = \dfrac{\cos\phi\cos\theta u_1}{m} \end{cases} \tag{47}$$

From Newton's second law, the total thrust in the control input is given by:

$$u_1 = F = m\sqrt{(U_1)^2 + (U_2)^2 + (U_3 + g)^2} \tag{48}$$

For the attitude angle ϕ, from the second line of (38), we obtain: $\sin\phi = \dfrac{-mU_2}{u_1}$. Thus, we can solve for the desired attitude angle:

$$\phi_d = \arcsin\left(\frac{-mU_2}{u_1}\right) \tag{49}$$

For the pitch angle θ, from the first and third lines of (38), we obtain: $\tan\theta = \dfrac{mU_1}{m(U_3 + g)}$. Thus, the desired pitch angle θ_d can be solved as:

$$\theta_d = \arctan\left(\frac{U_1}{U_3 + g}\right) \tag{50}$$

Table 1
Quadrotor UAV parameters.

Parameters	Value
m	2kg
l	0.2m
$I_{xx} = I_{yy}$	$0.06\text{kg} \cdot \text{m}^2$
I_{zz}	$0.01\text{kg} \cdot \text{m}^2$
$k_x = k_y = k_z$	0.01
$k_\phi = k_\theta = k_v$	0.012
g	9.8m/s
T	0.01s
ϵ	0.05

In summary, from (46), the total thrust u_1 and the desired attitude angle ϕ_d, θ_d can be derived. The attitude controller tracks the desired angle and couples it with the total thrust to achieve position control, thereby completing the system's closed-loop operation. □

6. Simulation

6.1. Simulation parameters

To validate the effectiveness and advantages of the control strategy proposed in this paper, a system model of the quadrotor UAV is built on the Simulink platform in MATLAB. The parameters of the quadrotor UAV are as follows:

By substituting the parameters from Table 1 into the state (2) of the quadrotor UAV continuous system, the matrices of the continuous system are obtained as follows:

$$\bar{A}_{1,2,3} = \begin{bmatrix} 0 & 1 \\ 0 & -0.005 \end{bmatrix}, \bar{A}_{4,5} = \begin{bmatrix} 0 & 1 \\ 0 & -0.04 \end{bmatrix},$$

$$\bar{A}_6 = \begin{bmatrix} 0 & 1 \\ 0 & -0.24 \end{bmatrix}, \bar{B}_{1,2,\ldots,6} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \bar{F}_{1,2,\ldots,6} = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

In this simulation, a sampling frequency of 0.01 is used, and the discrete system matrix of the above continuous system is calculated in the delta operator framework as follows:

$$A_{1,2,3} = \begin{bmatrix} 0 & 1.0000 \\ 0 & -0.0050 \end{bmatrix}, A_{4,5} = \begin{bmatrix} 0 & 0.9998 \\ 0 & -0.0400 \end{bmatrix},$$

$$A_6 = \begin{bmatrix} 0 & 0.9998 \\ 0 & -0.2397 \end{bmatrix}, B_{1,2,3} = F_{1,2,3} = \begin{bmatrix} 0.0050 \\ 1.0000 \end{bmatrix},$$

$$B_{4,5} = F_{4,5} = \begin{bmatrix} 0.0050 \\ 0.9998 \end{bmatrix}, B_6 = F_6 = \begin{bmatrix} 0.0050 \\ 0.9988 \end{bmatrix},$$

In this simulation, the UAV flies for 100 s. The x-axis trajectory is set as $\sin(2\pi t)$, the y-axis trajectory as $\cos(2\pi t)$, and the z-axis trajectory as $0.5t$. The UAV begins to fly under ideal conditions and is affected by slow varying disturbance, abrupt disturbance, and actuator failure at 20 s, 40 s, and 80 s respectively. The specific scenario is as follows.

Case 1. At 20 s, an oscillation cycle is applied to simulate a slow varying disturbance.

Case 2. At 40 s, a step disturbance is applied to simulate abrupt disturbance.

Case 3. At the 80-second mark, the actuator experiences partial failure along with drift or lock-up conditions.

During these times, the disturbance and input loss are modeled and simulated. The influence of control failure on the x-axis and y-axis is relatively small and can be ignored. The external disturbance and input loss affect the system as follows:

$$d_{1,2,3,4,5,6} = \begin{cases} 0, & 0 \le t < 20 \\ 0.5\sin(\pi t), & 20 \le t < 40 \\ 2, & 40 \le t \le 100 \end{cases} \tag{51}$$

$$\rho_{1,2,3,4} = \begin{cases} 1, & 0 \le t < 80 \\ 1 - 0.05(t - 80), & 80 \le t < 90 \\ 0.5, & 90 \le t \le 100 \end{cases}. \tag{52}$$

$$f_{1,2,3,4} = \begin{cases} 0, & 0 \le t < 80 \\ 0.4, & 80 \le t \le 100 \end{cases} \tag{53}$$

The control strategy proposed in this paper includes an adaptive estimator, an attitude controller, and a position controller. The parameters of the adaptive estimator are shown in Tables 2–4 below.

Table 2
Parameters of the adaptive estimator.

Parameters	Value
L_i^{j0}	$\begin{bmatrix} 100 & 0 \\ 0 & 100 \end{bmatrix}$
$\gamma_i^1 = \gamma_i^2$	5
γ_i^3	2
λ_i^j	0.1

Table 3
Parameters of the attitude controller.

Parameters	Value
$C_{41} = C_{51}$	8
C_{61}	10
$C_{42} = C_{52} = C_{62}$	10
$C_{43} = C_{53} = C_{63}$	0.2
$\alpha_4 = \alpha_5 = \alpha_6$	1.5
$\beta_4 = \beta_5 = \beta_6$	0.6
$\Gamma_{41} = \Gamma_{51} = \Gamma_{61}$	0.5
$\Gamma_{42} = \Gamma_{52} = \Gamma_{62}$	0.05

Table 4
Parameters of the position controller.

Parameters	Value
$C_{11} = C_{21}$	3.2
C_{31}	1.5
$C_{12} = C_{22}$	0.09
C_{32}	0.2
$C_{13} = C_{23}$	0.8
C_{33}	0.2
$\alpha_1 = \alpha_2$	0.6
α_3	0.8
$\beta_1 = \beta_2$	0.6
β_3	0.3
$\Gamma_{11} = \Gamma_{21} = \Gamma_{31}$	0.5
$\Gamma_{12} = \Gamma_{22} = \Gamma_{32}$	0.05

6.2. Simulation results

To validate the advantages of the control strategy proposed in this paper, a comparison is made between the tracking of the desired trajectory using three strategies: nonlinear recursive sliding mode control with adaptive disturbance estimator (NRSMC + ADE), nonlinear recursive sliding mode control with fixed-gain disturbance estimator (NRSMC + FDE), and traditional sliding mode control with adaptive estimator (SMC + ADE).

From Figs. 2 and 3, it can be seen that the NRSMC + ADE method has the smallest tracking error in all six channels, converges within ten seconds in the initial state, and exhibits the smoothest convergence process. The other two methods have slower convergence speeds compared to NRSMC + ADE and exhibit oscillations. At 20 s and 40 s, when affected by disturbances, NRSMC + ADE converges within three seconds, while the SMC + ADE method struggles to adapt to abrupt disturbances and requires longer time to converge. The NRSMC + FDE method fails to handle controller parameter adjustments effectively, leading to performance degradation and oscillations. At 80 s, when the system faces actuator failure, NRSMC + ADE is minimally affected due to its regulation capability, while the other methods show larger oscillations and errors.

From Figs. 4 and 5, it can be observed that the fixed-gain disturbance estimator (FDE) exhibits severe oscillations after facing abrupt disturbances at 40 s, while the adaptive disturbance estimator (ADE) is almost unaffected. At 80 s, due to input loss caused by actuator failure, FDE continues to oscillate, while ADE remains smooth and maintains minimal errors.

Based on the above analysis, all three methods can achieve fault-tolerant control of the UAV. However, compared with the other two methods, ADE + NRSMC has the fastest convergence speed and the smallest mean square error, offering advantages in both accuracy and rapidity. Moreover, NRSMC + ADE has the smallest overshoot, indicating its excellent transient stability and disturbance rejection capability. When estimating disturbances in x, y, z, directions, and the ϕ and θ angles, ADE is more stable and converges faster than FDE. For the ψ angle disturbance, both estimators demonstrate good performance. Overall, ADE outperforms FDE in terms

Table 5
Comparisons of root mean squared error.

Channel	NRSMC + ADE	NRSMC + FDE	SMC + ADE
x (m)	0.2831	1.3439	0.3950
y (m)	0.2061	0.2267	0.4988
z (m)	0.1121	0.9407	0.4988
ϕ (°)	0.0323	0.0523	0.0778
θ (°)	0.0344	0.1977	0.1156
ψ (°)	0.0013	0.0077	0712

Table 6
Comparisons of overshoot.

Channel	NRSMC + ADE	NRSMC + FDE	SMC + ADE
x (m)	10.2%	14.6%	10.8%
y (m)	12.1%	18.5%	26.2%
z (m)	3.4%	3.6%	3.7%

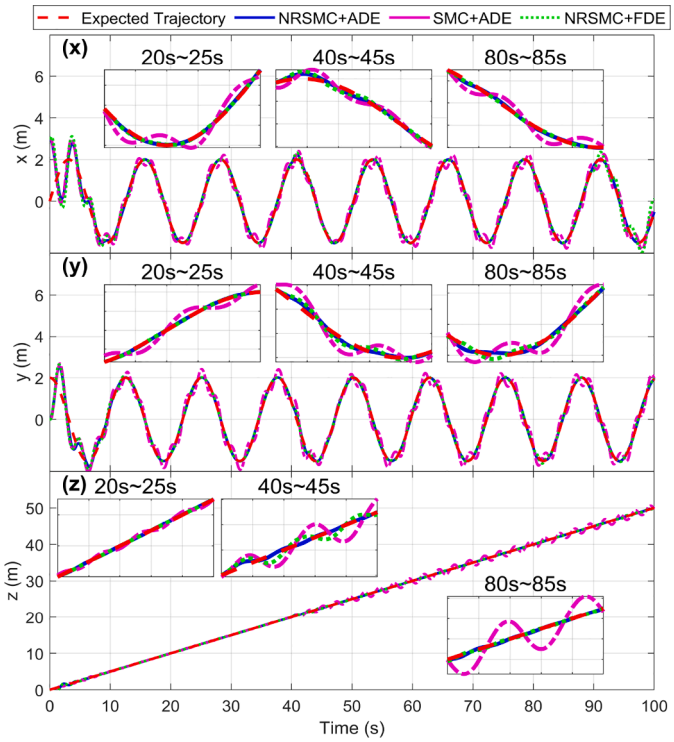

Fig. 2. Tracking of the trajectory.

of convergence speed, mean squared error, and smoothness. A detailed performance comparison can be found in the quantitative analysis in Tables 5 and 6.

From Fig. 6, it can be observed that during the initial trajectory tracking phase and when subjected to disturbances or actuator failure, the control input of the ADE + NRSMC exhibits slight high-frequency chattering, but it quickly converges. The comparison of the NRSMC + ADE's control inputs' performance with existing work is presented in the subsequent discussion section.

From Figs. 7 and 8, it can be seen that the first-level sliding surfaces of the position controller and attitude controller both exhibit good convergence performance, being able to converge to zero within a finite time. The second-level sliding surface shows chattering during the initial stage and when affected by disturbances or actuator failures, but it can achieve synchronous convergence and becomes smooth after convergence.

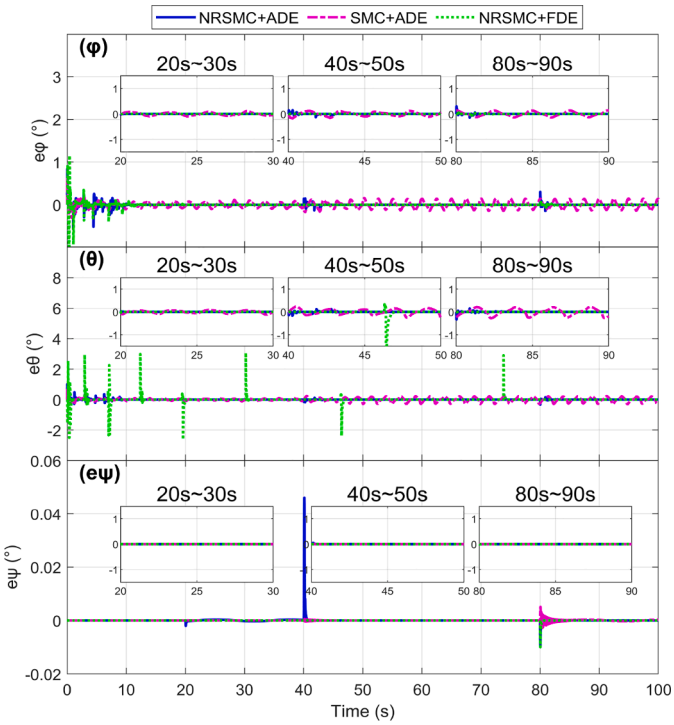

Fig. 3. Tracking error of the attitude.

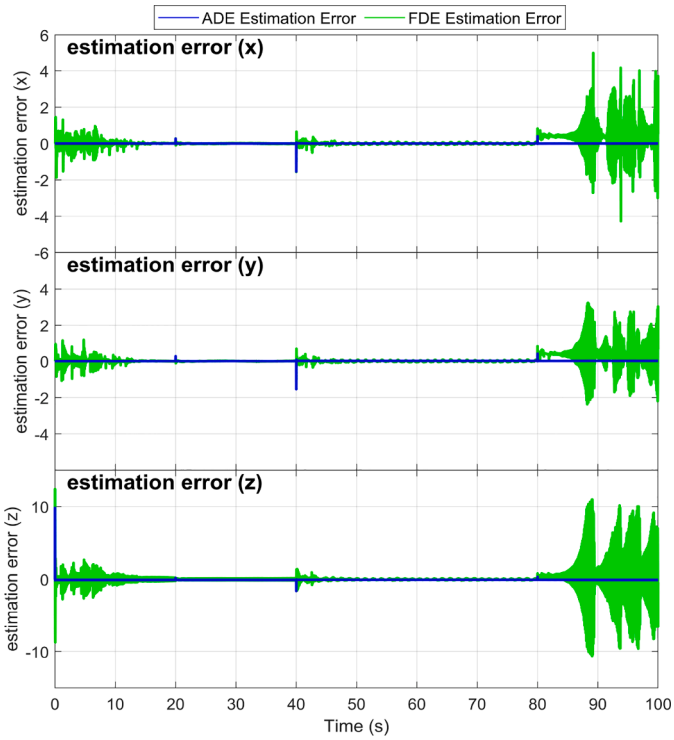

Fig. 4. Tracking of the position disturbance.

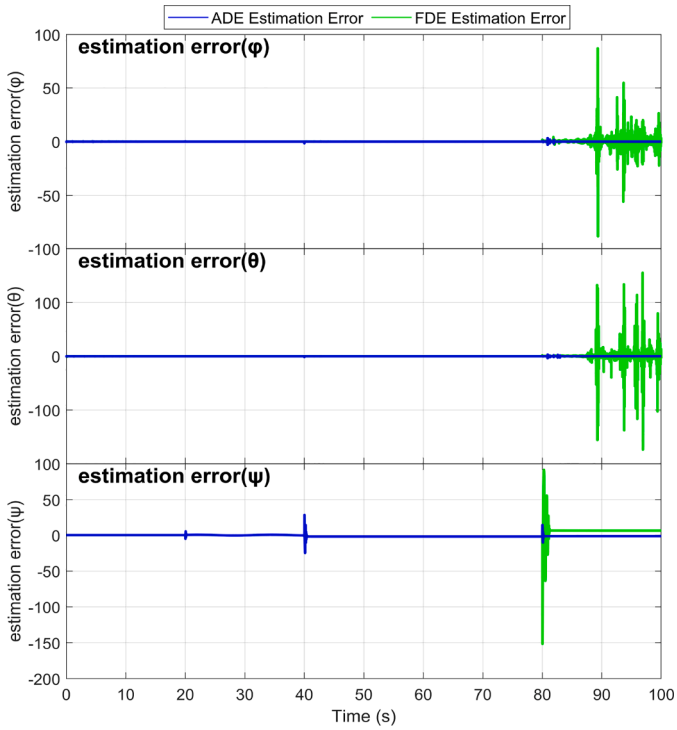

Fig. 5. Tracking of the angle disturbance.

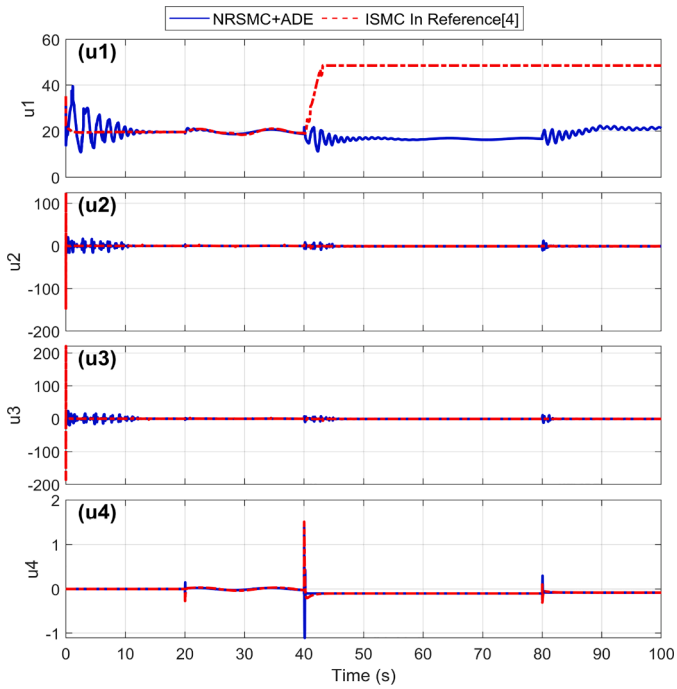

Fig. 6. Comparison of the control input.

7. Discussion

It is worth noting that the control input amplitude of the NRSMC + ADE approach is significantly lower compared to traditional sliding mode control methods. After convergence, the amplitude becomes extremely small, generating smooth and efficient control signals for the majority of the operational time.

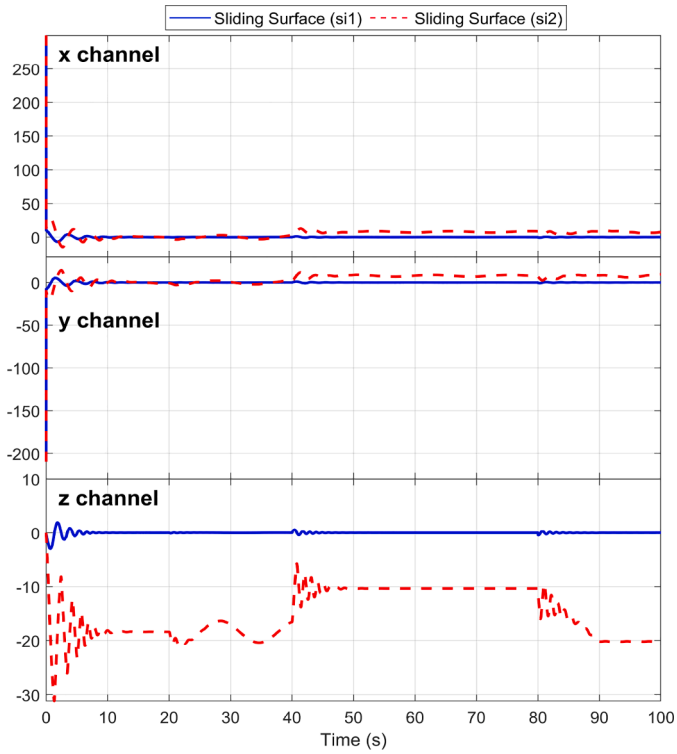

Fig. 7. Sliding surfaces of position controller.

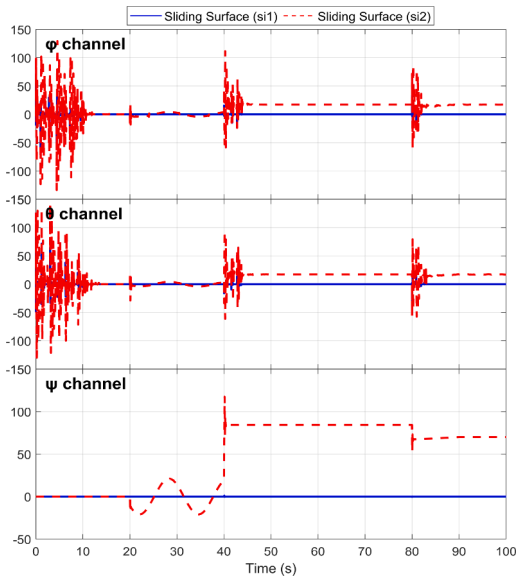

Fig. 8. Sliding surfaces of attitude controller.

Compared with the model in [3], on one hand, (1) explicitly encompasses both multiplicative and additive faults. This modeling approach can describe a broader range of fault modes (such as compound faults where partial loss of actuator effectiveness and constant deviation occur simultaneously), providing a more universal theoretical foundation for designing fault-tolerant controllers capable of handling complex fault scenarios. On the other hand, the study in [3] achieved good results in attitude tracking, whereas the method presented in this paper performs well in both attitude and position tracking.

In the simulation environment set up in this paper, compared with the control method in [4], the control input u_1 of the NRSMC + ADE has a lower steady-state value after receiving disturbances, but exhibits larger initial oscillations. The ISMC is ini-

tially smooth but has a higher steady-state value. For control inputs u_2 and u_3, ISMC initially has larger control inputs. For u_4, the performance of both is comparable. Overall, although NRSMC + ADE's control inputs show slight chattering in the initial stage and when facing disturbances and actuator faults, its steady-state values are lower than those of the ISMC, which allows the UAV motors to have a longer lifespan under abrupt disturbances and complex actuator faults.

The control strategy proposed in this paper, due to the use of an estimator, has good compensation capability for uncertainties in the UAV model and model parameters.

8. Conclusion

This paper addresses the fault-tolerant control problem for quadrotor UAVs and proposes a control strategy that integrates an Adaptive Disturbance Estimator (ADE) with Nonlinear Recursive Sliding Mode Control (NRSMC) within the delta operator framework. Simulation results demonstrate that, compared with existing studies, the proposed method ensures stable and high-precision trajectory tracking for the UAV in the presence of abrupt disturbances and actuator faults. Moreover, the control input remains moderate during steady-state operation, effectively handling abrupt disturbances and compound actuator faults while guaranteeing actuator longevity.

However, the controller proposed in this paper is only applicable to the control of a single UAV, and future work could attempt to extend it to formation control.

CRediT authorship contribution statement

Shuo Gao: Writing – original draft, Conceptualization; **Bo-Chao Zheng:** Writing – review & editing, Supervision, Methodology; **Hanqing Qu:** Software, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] F.A. Aliabad, H. Ghafarian, A. Malamiri, A. Sarsangi, E. Sekertekin, Ghaderpour, Identifying and monitoring gardens in urban areas using aerial and satellite imagery, Remote Sens. 15 (16) (2023) 4053.
[2] B. Mendu, N. Mbuli, State-of-the-art review on the application of unmanned aerial vehicles (UAVs) in power line inspections: current innovations, trends, and future prospects, Drones 9 (4) (2025) 265.
[3] V.T. Mai, F. Jani, K.A. Alattas, E. Ghaderpour, A. Mohammadzadeh, A robust finite-time fault-tolerant tracking control for quadrotor attitude system with stochastic actuator faults and input delays, IEEE Access 13 (2025) 64627–64637.
[4] S. Zhou, K. Guo, X. Yu, L. Guo, L. Xie, Fixed-time observer based safety control for a quadrotor UAV, IEEE Trans. Aerosp. Electron. Syst. 57 (5) (2021) 2815–2825.
[5] K. Liu, R. Wang, S. Zheng, S. Dong, G. Sun, Fixed-time disturbance observer-based robust fault-tolerant tracking control for uncertain quadrotor UAV subject to input delay, Nonlinear Dyn. 107 (3) (2022) 2363–2390.
[6] R.C. Avram, X. Zhang, J. Muse, Quadrotor actuator fault diagnosis and accommodation using nonlinear adaptive estimators, IEEE Trans. Control Syst. Technol. 25 (6) (2017) 2219–2226.
[7] M. Mazare, M. Taghizadeh, P. Ghaf-Ghanbari, E. Davoodi, Robust fault detection and adaptive fixed-time fault-tolerant control for quadrotor UAVs, Rob. Auton. Syst. 179 (2024) 104747.
[8] S. Ranjan, S. Majhi, Fixed-time state observer-based robust adaptive neural fault-tolerant control for a quadrotor unmanned aerial vehicle, Int. J. Adapt. Control Signal Process. 39 (1) (2025) 132–151.
[9] B.H. Du, A. Polyakov, G. Zheng, Q. Quan, Quadrotor trajectory tracking by using fixed-time differentiator, Int. J. Control 92 (12) (2018) 2854–2868.
[10] L. Cui, X. Hou, Z. Zuo, H. Yang, An adaptive fast super-twisting disturbance observer-based dual closed-loop attitude control with fixed-time convergence for UAV, J. Franklin Inst. 359 (6) (2022) 2514–2540.
[11] L. Wang, J. Chen, Y. Huang, Y. Zhu, H. Pei, Geometric attitude tracking control of quadrotor UAVs with adaptive extended state observers, ISA Trans. 160 (2025) 289–297.
[12] S. Ranjan, S. Majhi, Fixed-time observer-based adaptive free-will arbitrary time intelligent fault-tolerant control for an autonomous quadrotor, Int. J. Syst. Sci. 56 (14) (2025) 3404–3427.
[13] H. Yang, Y. Xia, P. Shi, Observer-based sliding mode control for a class of discrete systems via delta operator approach, J. Franklin Inst. 347 (7) (2010) 1199–1213.
[14] D. Ginoya, P.D. Shendge, S.B. Phadke, Delta-operator-based extended disturbance observer and its applications, IEEE Trans. Ind. Electron. 62 (9) (2015) 5817–5828.
[15] Y. Wu, B. Zheng, Research on Attitude Control of Quadrotor, Technical Report, Nanjing University of Information Science and Technology, 2022.
[16] C. Cao, C. Wei, Y. Liao, Y. Zhang, J. Li, On novel trajectory tracking control of quadrotor UAV: a finite-time guaranteed performance approach, J. Franklin Inst. 359 (16) (2022) 8454–8483.
[17] M. Ebrahimpour, M. Lungu, M. Kakavand, Antisaturation fixed-time backstepping fuzzy integral sliding mode control for automatic landing of fixed-wing unmanned aerial vehicles, J. Franklin Inst. 361 (17) (2024) 107185.
[18] S. Lian, Y. Zhu, W. Meng, K. Shao, H. Li, Adaptive fault-tolerant control for quadrotor based on the second-order fast nonsingular terminal sliding mode control, IEEE Trans. Ind. Electron. 72 (8) (2025) 8322–8332.
[19] W. Yu, J. Yan, X. Pan, S. Tan, H. Cao, Y. Song, Fault-tolerant attitude tracking control with practical finite time convergence for unmanned aerial vehicles under actuation faults, Int. J. Robust Nonlinear Control 32 (6) (2022) 3737–3753.